

A Multi-region Logistics SaaS on NATS

How shippers and transporters meet through a governed marketplace; how NATS accounts isolate tenants; and how the same service and data model deploys across regions and logistics edge locations.

THE GOVERNING INVARIANT

Region is placement. Tenant is authority. Organisation is participation. A marketplace tenant may span regions and contain many shippers and transporters, but every interaction is mediated by platform services and constrained by account, membership, role, and subject permissions.

WHAT THE PUBLIC LINEBOOKER MODEL ADDS

The business loop is not merely "customer talks to transporter." It is **procure, aggregate, execute and settle**: fixed-lane tenders, bidding loads, truck commitment, route execution, live tracking, digital POD, claims and consolidated payment.

THE TEN ARCHITECTURE CARDS

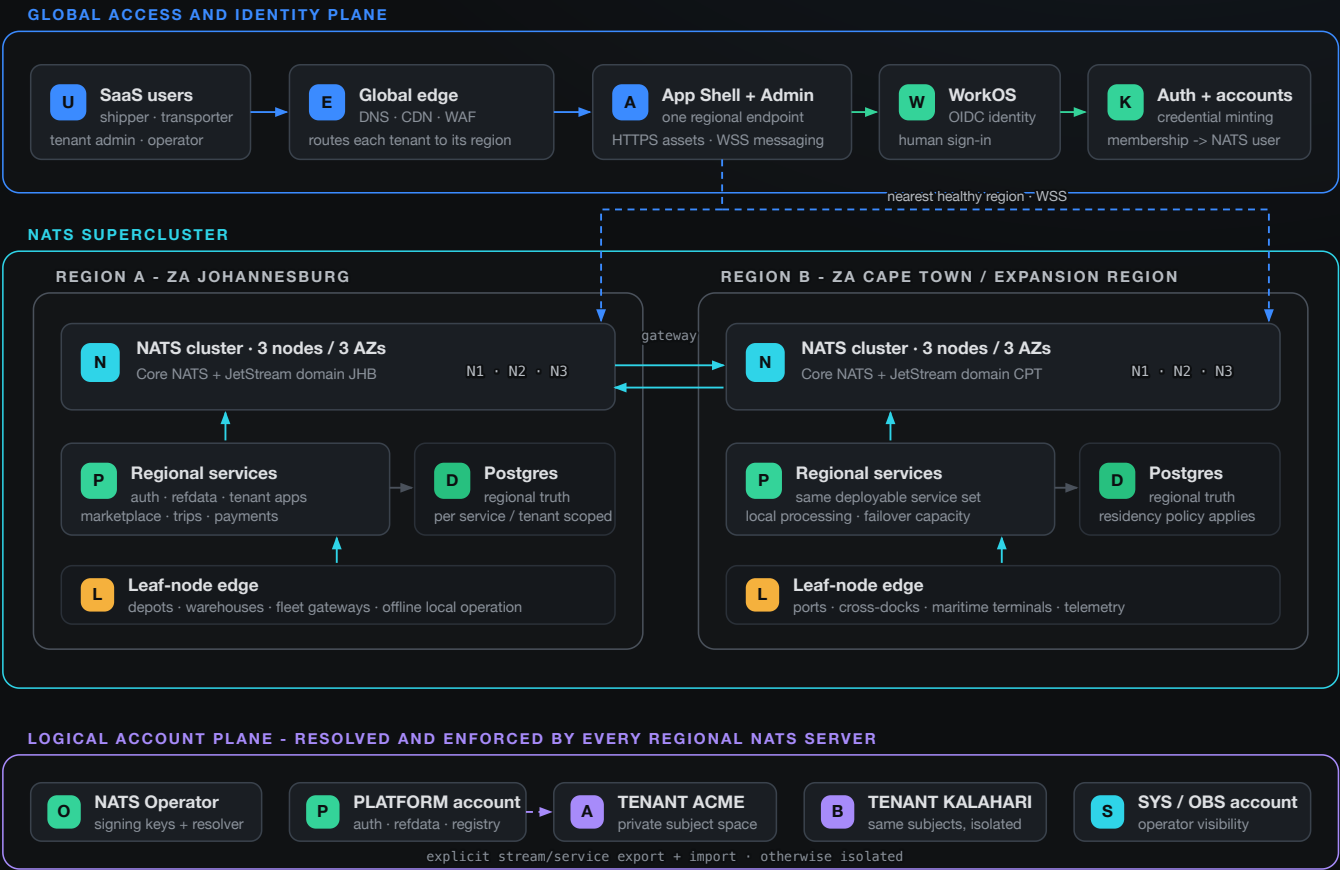
- 01 · System landscape
- 02 · Shipper-transporter engagement
- 03 · Service and bounded-context model
- 04 · Tenant, organisation and account model
- 05 · Human identity to NATS session
- 06 · Regional deployment model
- 07 · Data and event ownership
- 08 · Tracking and logistics edge
- 09 · Failure, recovery and residency
- 10 · Provisioning and operations

01

REGION IS PLACEMENT · ACCOUNT IS AUTHORITY

One SaaS, many regions, hard tenant walls

A regional NATS supercluster carries the live logistics fabric. Human identity enters through WorkOS; signed NATS credentials place every browser and service inside exactly one account boundary.



Solid arrow = request/read · Dashed arrow = pushed credential/event · Dotted = selective replication · Blue = client · Cyan = NATS · Green = platform · Purple = tenant

GEOGRAPHY

Clusters provide regional HA.
Gateways make the clusters one supercluster. A tenant can operate in several regions without region becoming part of its identity.

IDENTITY + TENANCY

WorkOS identifies the human; NATS authorizes the connection.
The minted credential binds a user or service to one account and a least-privilege subject set.

DATA PLACEMENT

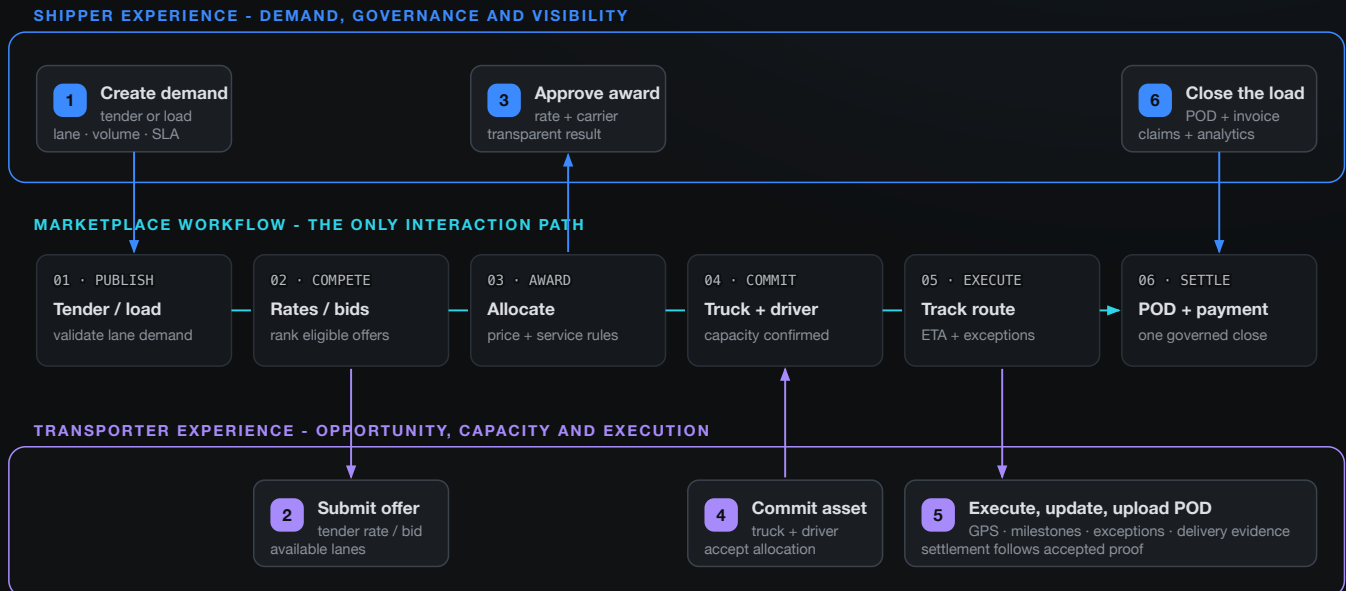
Persistence stays regional by policy. Postgres is authoritative; JetStream domains retain local event history. Mirror only the streams approved for DR or global views.

02

THE MARKETPLACE IS THE CONVERSATION

How shippers and transporters engage

They never receive a direct technical channel to one another. Each side acts through a role-specific client; marketplace services validate, rank, allocate, notify and preserve the audit trail.



Blue = shipper action · Purple = transporter action · Cyan = marketplace state transition · Every transition is authorised, persisted and observable

FIXED CONTRACT LANES

Longer-term tenders establish lane rates, committed volumes and selected transporter capacity.

DAILY BIDDING LOADS

Spot opportunities absorb excess fleet and return legs without bypassing vetting or allocation rules.

EXECUTION CLOSES THE LOOP

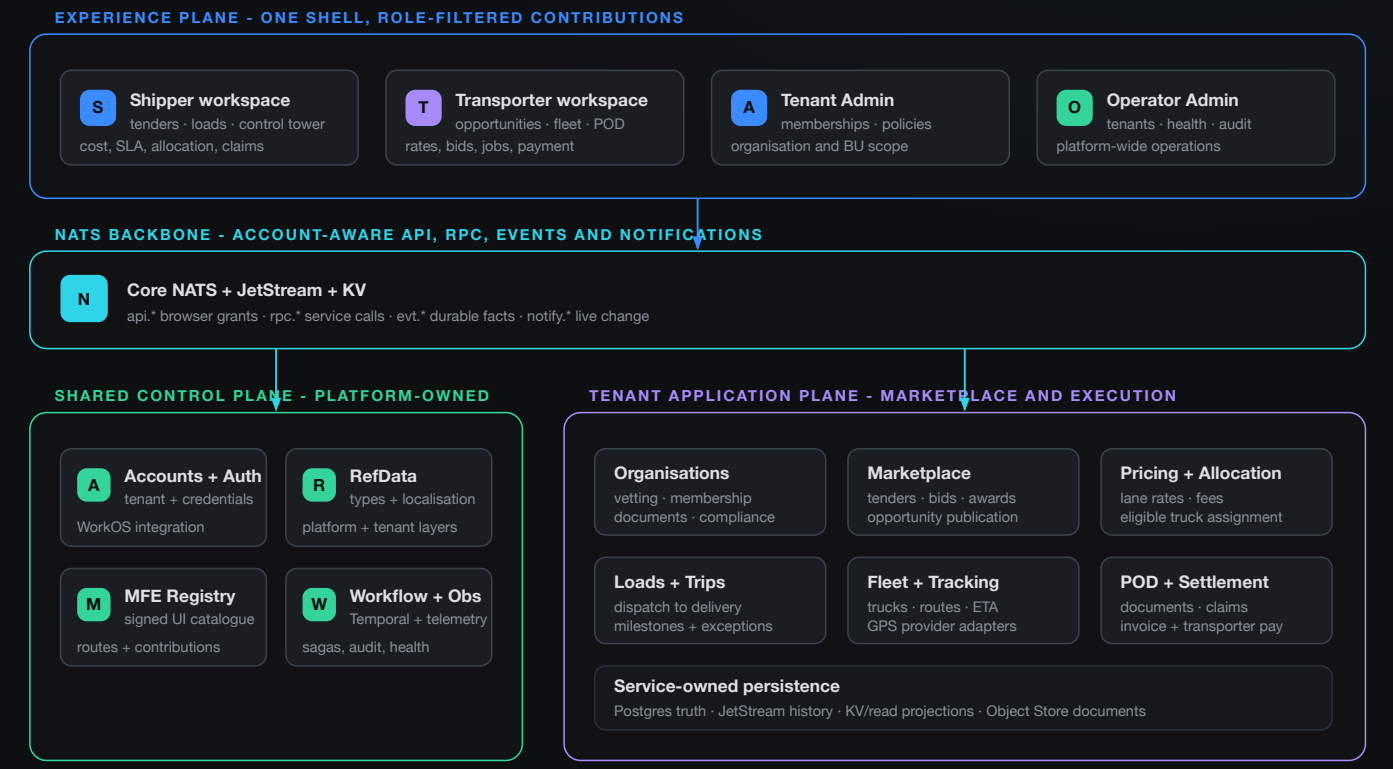
Tracking, exceptions, digital POD, claims and settlement remain part of the same governed load record.

03

CONTROL PLANE AROUND AN APPLICATION PLANE

Services follow business ownership

The boxes below are bounded contexts and deployable ownership units. They are not a mandate to create one container, database or team for every noun on day one.

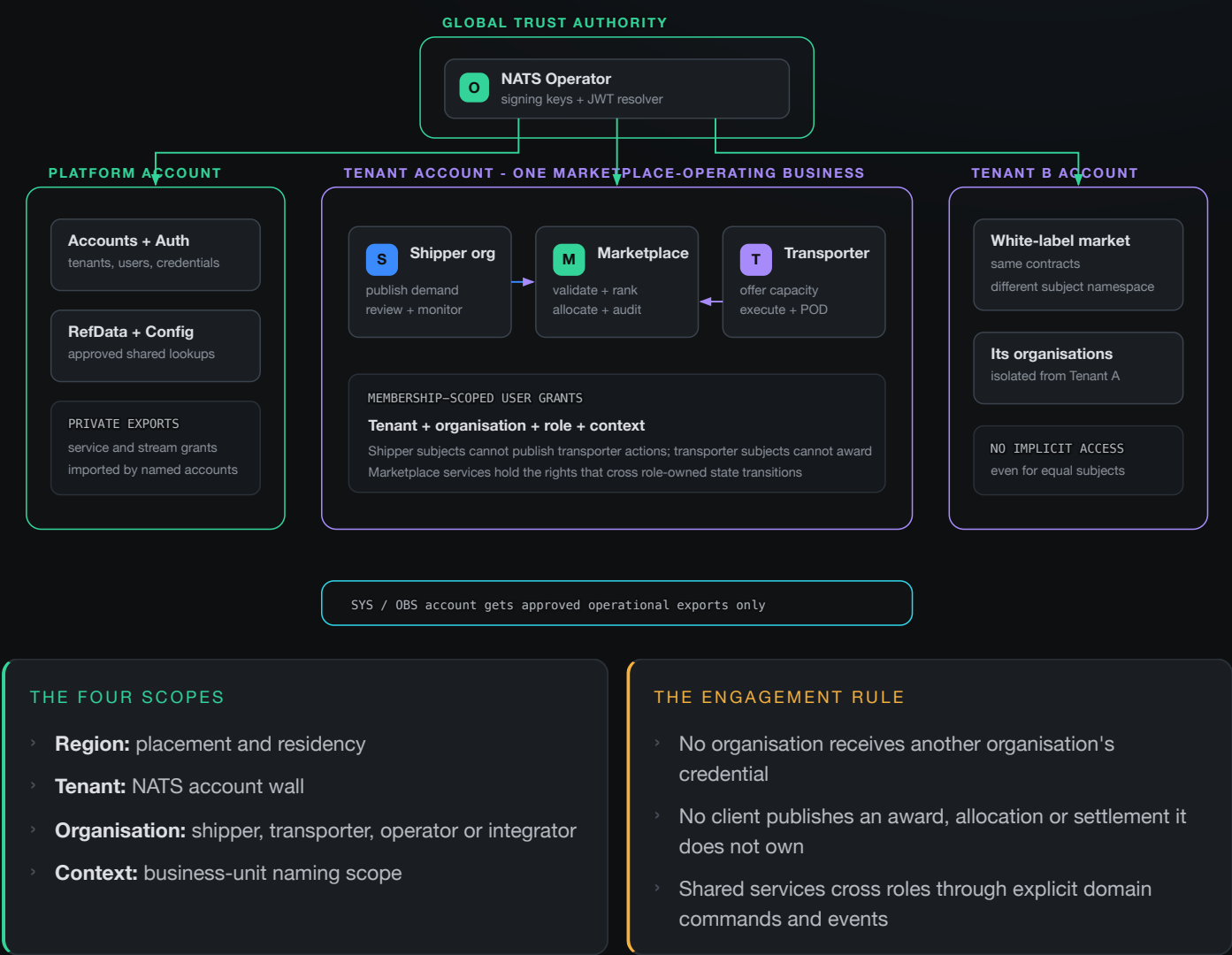


04

FOUR AXES, FOUR DIFFERENT JOBS

The tenant is the marketplace operator

A region is not a tenant, and a shipper or transporter is not automatically a tenant. They are organisations participating in a marketplace tenant through scoped memberships.

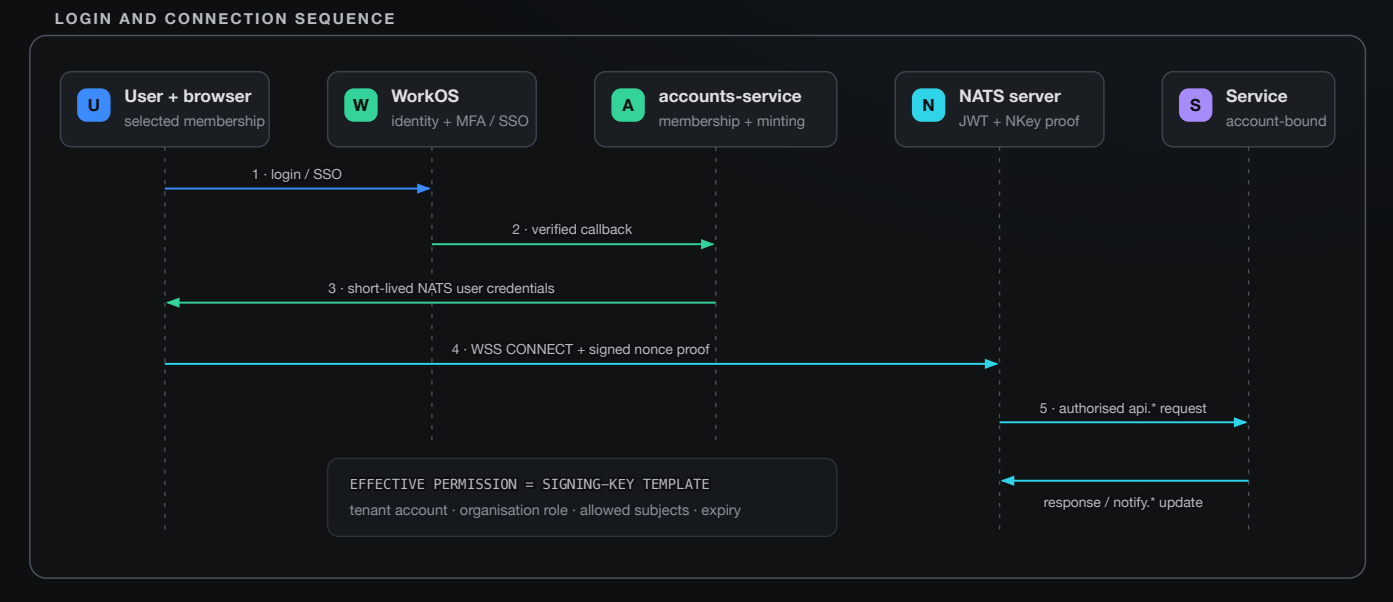


05

IDENTITY IS NOT MESSAGING AUTHORITY

From human sign-in to a NATS session

WorkOS establishes who the person is. Accounts-service decides which active membership they may use, then mints a short-lived credential whose signing key constrains the effective NATS permissions.



Blue = browser identity action · Green = identity and credential control · Cyan = authenticated NATS traffic · The browser never sends a tenant id that widens its credential

AUTHENTICATION

WorkOS handles password, enterprise SSO, MFA and directory-driven lifecycle.

AUTHORISATION

NATS validates the operator-account-user trust chain and enforces subject permissions server-side.

HTTP EXCEPTION

Where direct HTTP is unavoidable, first mint a short-lived, single-use capability over the authenticated NATS connection.

06

REPEATABLE REGIONAL CELLS

Deploy regions, not bespoke tenant stacks

Each region is a repeatable cell: a three-node NATS cluster, the same versioned service set, service-owned persistence and regional ingress. Gateways connect cells without turning the WAN into one cluster.



POOLED DEFAULT

- › One service deployment per region serves many tenant accounts
- › Each active tenant gets its own NATS connection and data scope
- › Best operational efficiency for the standard tier

OPTIONAL SILO TIER

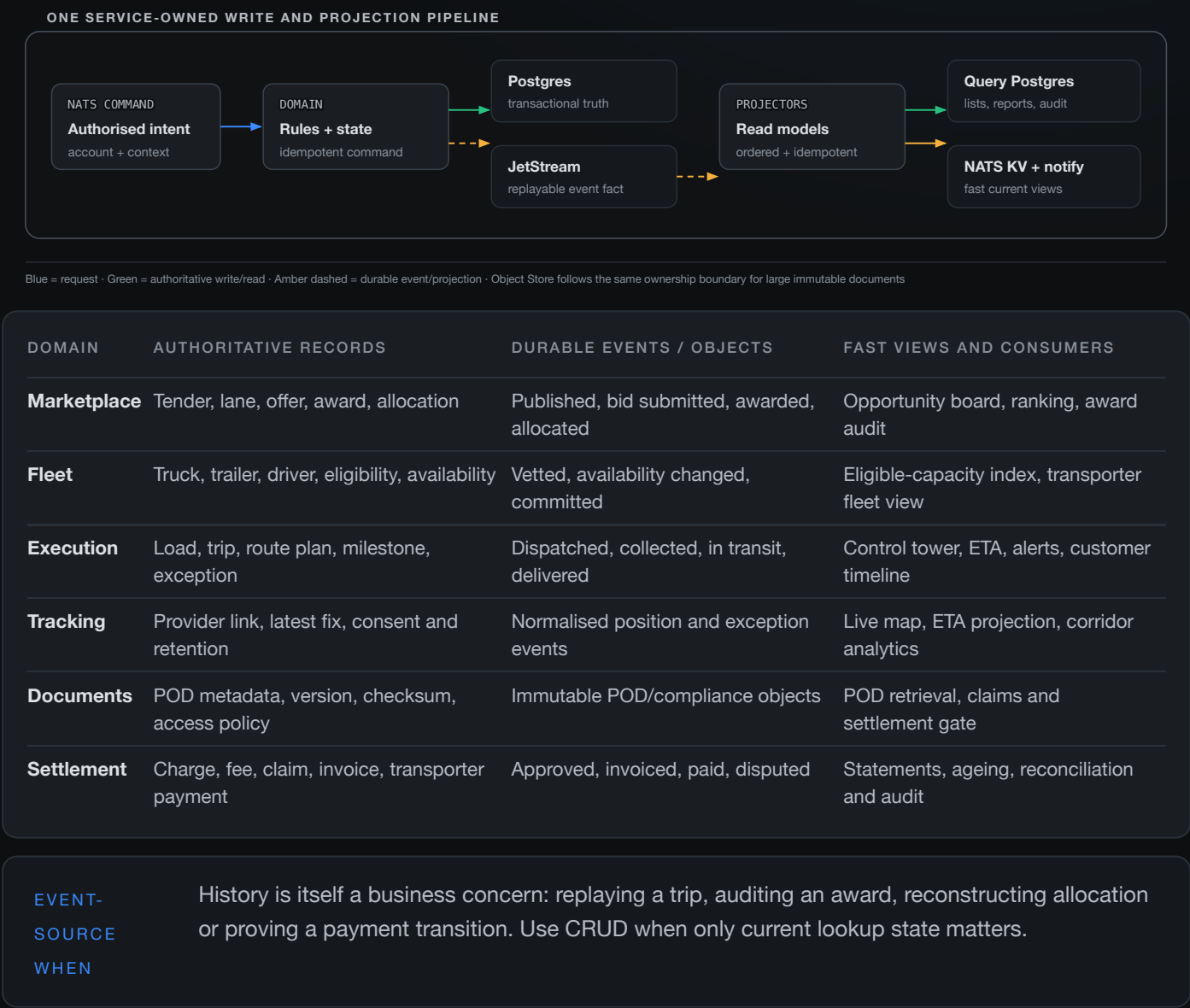
- › Dedicate selected service replicas or storage to a tenant
- › Keep the same subjects, contracts and onboarding control plane
- › Use for regulation, scale or noisy-neighbour isolation

07

TRUTH, HISTORY AND FAST VIEWS ARE DIFFERENT JOBS

Each service owns its facts

Postgres is authoritative for transactional state. JetStream retains replayable domain facts. KV and projections make current state fast to read; Object Store holds POD and compliance documents.

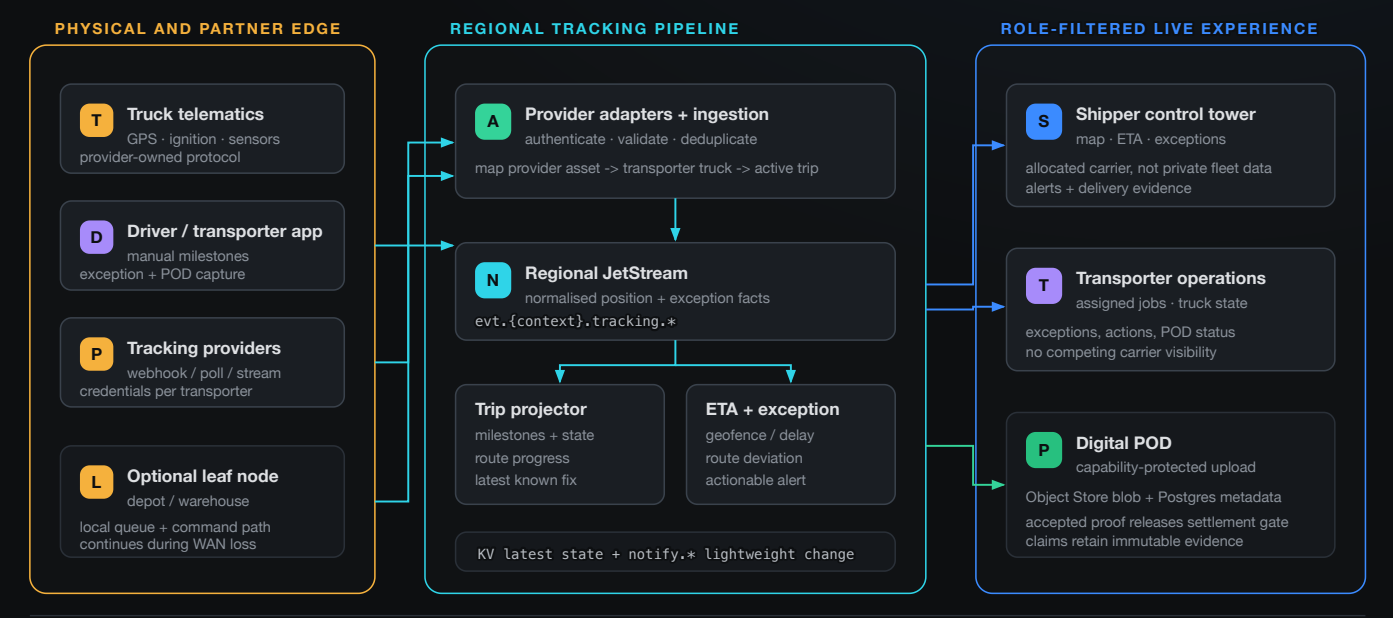


08

MANY TELEMETRY SHAPES, ONE TRIP TIMELINE

Tracking becomes a normalised business signal

Vehicle devices, transporter systems and driver apps differ. Provider adapters normalise their inputs before trip services derive milestones, ETA and exceptions for both client experiences.



PRIVACY BOUNDARY

Share only tracking for an allocated load and its permitted time window. A shipper never gains general access to a transporter's fleet.

QUALITY BOUNDARY

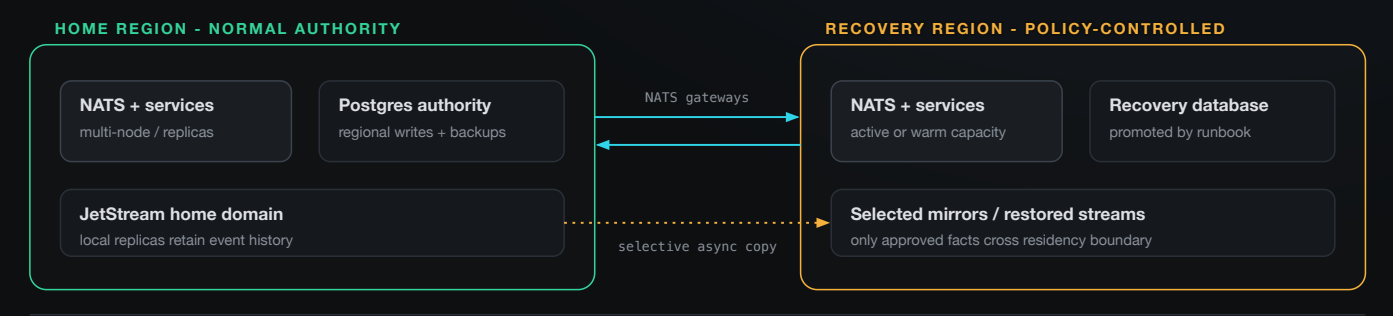
Reject stale, impossible or duplicate fixes; retain provider provenance so ETA and dispute decisions remain explainable.

09

AVAILABILITY DOES NOT ERASE RESIDENCY

Fail locally before failing across regions

A node or service failure should be invisible inside its regional cell. A regional evacuation is a policy-driven event: traffic may move quickly, but authoritative data moves only through an approved DR design.



FAILURE	EXPECTED RESPONSE	WHAT MUST NOT HAPPEN
One NATS node / AZ	Clients reconnect inside the cluster; JetStream leader changes	Cross-region failover for a local node loss
Service replica	Queue subscriber or load balancer selects a healthy replica	Duplicate non-idempotent command execution
WAN / gateway	Regions and leaf sites continue locally; replay after reconnection	Assume a disconnected edge is permanently failed
External GPS / WorkOS	Degrade the affected capability and surface stale/blocked state	Lose active trip state or widen access to compensate
Whole region	Apply tenant-specific routing, data-promotion and residency runbook	Promote unapproved or incomplete data globally

DECIDE CONTRACTUALLY

Home region, permitted recovery regions, RTO, RPO, POD residency and whether active trips may fail across borders.

TEST OPERATIONALLY

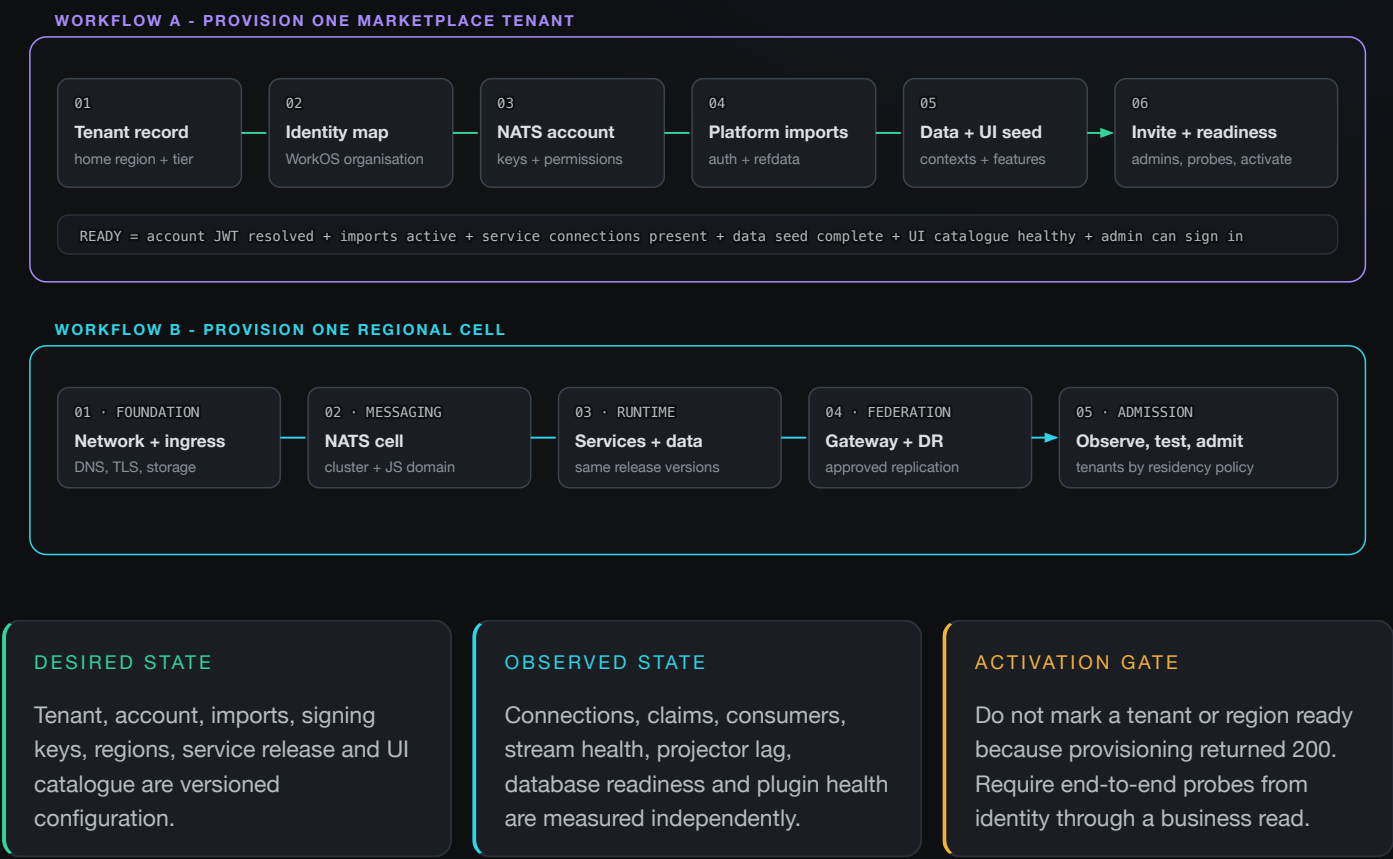
Node loss, account suspension, expired credentials, gateway partition, stale telemetry, projector replay and database promotion.

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TWO REPEATABLE CONTROL-PLANE WORKFLOWS

Add a tenant; clone a region

Neither operation is a ticket-driven collection of manual NATS edits. The control plane applies versioned policy, records every step and exposes readiness only when identity, messaging, data and UI agree.



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EVIDENCE AND DESIGN STATUS

What is sourced, settled and proposed

This document combines the repository's implemented NATS patterns, public statements about Linebooker's operating model and explicit architecture proposals for a future multi-region logistics SaaS.

REPOSITORY-GROUNDED

- › NATS operator mode with PLATFORM, tenant and system account boundaries
- › Accounts-service mints short-lived credentials; auth-service is merged into it
- › Per-tenant service connections; context-scoped subjects and KV keys
- › Postgres projection as truth with JetStream history and eager KV cache
- › WorkOS user lifecycle is documented, with implementation status called out separately

LINEBOOKER PUBLIC MODEL

- › Marketplace connects shippers/customers with vetted transport companies and trucks
- › Fixed contract lane rates dominate; bidding supports surplus capacity and return legs
- › Procure, aggregate and execute cover tendering, allocation, tracking, POD and payment
- › GPS-integrated tracking, digital POD, control-tower visibility and consolidated settlement
- › Transporters register, are vetted, receive opportunities, submit/accept, execute and get paid

ARCHITECTURE PROPOSALS REQUIRING DESIGN APPROVAL

- › Johannesburg and Cape Town shown as example regional cells; actual cloud regions and residency classes remain to be selected
- › Pooled regional services are the default, with an optional silo tier for regulation, scale or noisy-neighbour isolation
- › Tenant means marketplace-operating business; shipper and transporter organisations participate inside it through membership-scoped grants
- › Selected JetStream mirrors, Postgres recovery and Object Store replication are governed separately; no automatic global data movement is assumed
- › Domain boxes are bounded contexts; final service boundaries, event schemas, RTO/RPO and retention policies need business-rule approval

PRIMARY PUBLIC REFERENCES

Linebooker: [linebooker.com](#) · [Platform Solutions](#) · [For Transporters](#) · [FAQ](#)

NATS: [Adaptive Edge Architecture](#) · [Accounts](#) · [Gateways](#) · [Leaf Nodes](#)

AWS architecture framing: [SaaS Architecture Fundamentals](#) · [Architecting Multi-Region SaaS Solutions](#)

READING RULE

Public business statements describe outcomes, not internal implementation. Service names, NATS subjects, deployment tiers and failure behavior in these cards are the proposed technical model unless the repository already implements and documents them.